

CALIMESA FIRE DEPARTMENT
POLICY MANUAL
VOLUME 3
Policy: 301.17 Heat Illness Prevention Program

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Types of Heat Illness - continued

CONDITIONS	PROBABLE HEALTH RISK	POSSIBLE SYMPTOMS	TYPICAL ACTIONS
<u>Sun Burn</u>	Low	Skin becomes red, painful, and abnormally warm after sun exposure.	Consult a doctor if the following symptoms are present: 1. Fever 2. Fluid-filled blisters 3. Severe pain
<u>Heat Rash</u>	Low	Looks like a red cluster of pimples or small blisters. It is more likely to occur on the neck and upper chest, in the groin, under the breasts, and in elbow creases.	1. Seek a cooler, less humid environment. 2. Keep the affected area dry. 3. Dusting powder may be used to increase comfort, but avoid using ointments or creams because they keep the skin warm and moist and may make the condition worse. 4. Treating heat rash is simple and usually does not require medical assistance.
<u>Heat Fatigue</u>	Low	May feel too tired or too weak to concentrate, an early warning sign of heat stress.	Increase fluid intake and take a break.
<u>Heat Syncope</u>	Low	Person may feel faint or feel lightheaded. The skin is pale, cool and moist. There is a feeling of lightheadedness when the person changes position, such as moving from a squatting position to a standing position.	1. Recovery is rapid once the person lies down in a cool environment. 2. Slowly drink water, clear juice, or a sports beverage. WARNING: If on a low-salt diet, discuss drinking sports beverages with a doctor.

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CONDITIONS	PROBABLE HEALTH RISK	POSSIBLE SYMPTOMS	TYPICAL ACTIONS
<u>Heat Cramps</u>	Somewhat serious	Muscle cramps (usually in the abdomen, arms or legs) due to excessive sweating.	1. Stop all activity and sit quietly in a cool place. 2. Drink clear juice or a sports beverage. WARNING: If on a low-salt diet, discuss drinking sports beverages with a doctor. 3. Do <u>not</u> return to strenuous activity for a few hours after the cramps subside because further exertion may lead to heat exhaustion or heat stroke, very serious conditions. 4. See medical attention for heat cramps if they do not subside in 1 hour, and/or if the worker has heart problems, and/or the worker is on a low-sodium diet.
<u>Heat Exhaustion</u>	More serious	Pale or flushed appearance, weakness, sweating, headache, moist, clammy skin, dizziness, heart problems or high blood pressure. sometimes nausea or slight fever, muscle cramps. Skin may be cool and moist. The person's pulse rate will be fast and weak, and breathing will be fast and shallow. The person might have a low-grade fever, generally less than 104° F. If untreated, it can progress to heat stroke.	1. Get medical help immediately if any of the symptoms are severe and/or the victim has problems or high blood pressure. 2. Meanwhile, move the person to a cool, shaded, or air conditioned place. a. Loosen clothing. b. Have the person slowly drink cool, nonalcoholic fluids, not iced or a sports drink containing electrolytes. c. Have the person take a cool shower d. Or apply cool compresses to the body or spray him or her with cool water along with fanning. e. Try to get them into an air-conditioned environment. f. Elevate feet 8 - 12 inches. g. Allow the person to rest. 3. Seek medical attention if symptoms worsen or last longer than 1 hour.

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CONDITIONS	PROBABLE HEALTH RISK	POSSIBLE SYMPTOMS	TYPICAL ACTIONS
<u>Heat Stroke</u>	True life-threatening emergency	An extremely high body temperature (above 103° F., orally), red, hot and dry skin (no sweating), hallucinations, chills, slurred speech, rapid, strong pulse, throbbing headache, dizziness, nausea, confusion, unconsciousness.	1. Call for medical help immediately (911). 2. Meanwhile, get the victim to a cool, shady area. 3. Cool the person down rapidly using whatever methods are available to include spraying the person with cool water from a garden hose; sponging the person with cool water. If the humidity is low, wrap the victim in a cool, wet sheet and fan the person vigorously. 4. Monitor the ABCs (airway, breathing and circulation). 5. Monitor body temperature and continue cooling efforts until the body temperature drops to 101° to 102° F. 6. Do <u>not</u> give the person fluids to drink. 7. If emergency medical personnel are delayed, call the emergency room (ER) for further instructions. 8. Sometimes a victim's muscles will begin to twitch uncontrollably as a result of heat stroke. If muscles begin to twitch uncontrollably: a. Keep the victim from injuring him or herself. b. Do <u>not</u> place any object in the mouth. c. Do <u>not</u> give fluids. 9. If there is vomiting, make sure the airway remains open by turning the victim on his or her side.

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Preventive actions include:

- Increase your fluid intake, regardless of your activity. Water is the predominant fluid you drink. To replenish salts and minerals that are lost through sweating, an occasional drink of a clear juice or sports drink will help. Drink one quart of water per hour per employee for the entire shift. WARNING: If your doctor generally limits the amount of fluid you drink or has you on water pills, ask the doctor how much you should drink when working in hot weather.
- Do not drink liquids that contain alcohol, or large amounts of sugar. These drinks actually cause you to lose more body fluid. Also avoid icy cold drinks because they can cause stomach cramps.
- Limit activity in an extremely hot environment. Adjust work schedules so when possible, outdoor work is completed in the morning.
- Watch others for symptoms of heat-related illness. Older persons are particularly vulnerable.
- Wear sunscreen on unprotected areas of the skin. Sun screen should be SPF 15 or higher (*broad spectrum or UVA/UVB protection*)
- Wear lightweight, light colored, loose fitting, long-sleeved, long-legged clothing, and a wide-brimmed hat if possible.
- Ask your doctor whether medications you take affect your body's response to the heat.
- If exertion in the heat makes your heart pound and leaves you gasping for breath, STOP all activity. Get into a cool area or at least into the shade, and rest, especially if you become lightheaded, confused, weak or faint. Call for help from a fellow employee or supervisor.
- Avoid hot foods and heavy meals; they add heat to your body.

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3. Definitions

- a. **Acclimatization** means temporary adaptation of the body to work in the heat that occurs gradually when a person is exposed to it. Acclimatization peaks in most people within four to fourteen days of regular work for at least two hours per day in the heat.
- b. **Heat illness** means a group of serious medical conditions resulting from the body's inability to cope with a particular heat load and includes heat cramps, heat exhaustion, heat syncope and heat stroke. In this Program, sunburn and heat rash are also included because such problems also emanate from exposure to the sun and/or heat.
- c. **Environmental risk factors for heat illness** means working conditions that create the possibility that heat illness could occur, including temperature, relative humidity, radiant heat from the sun and other sources, conductive heat sources such as the ground, air movement, workload severity and duration, protective clothing and personal protective equipment worn by employees.
- d. **Personal risk factors for heat illness** mean factors such as an individual's age, degree of acclimatization, health, water consumption, alcohol consumption, caffeine consumption, and use of prescription medications that affect the body's water retention or other physiological responses to heat.
- e. **Recovery period** means a period of time to recover from the heat in order to prevent heat illness.
- f. **Shade** means blockage of direct sunlight. One indicator that blockage is sufficient is when objects do not cast a shadow in the area of blocked sunlight. Shade is not adequate when heat in the area of shade defeats the purpose of shade, which is to allow the body to cool. For example, a car sitting in the sun does not provide acceptable shade to a person inside it, unless the car is running with air conditioning. Shade may be provided by any natural or artificial means that does not expose employees to unsafe or unhealthy conditions. Canopies, umbrellas and other temporary structures or devices may be used to provide shade.

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- g. **Temperature** means the dry bulb temperature in degrees Fahrenheit obtainable by using a thermometer to measure the outdoor temperature in an area where there is no shade. While the temperature measurement must be taken in an area with full sunlight, the bulb or sensor of the thermometer should be shielded (for example, with the hand or some other object) from direct contact by sunlight while taking the measurement.

4. **Heat Index**

Our bodies dissipate heat by varying the rate and depth of blood circulation, by losing water through the skin and sweat glands, and as a last resort, by panting, when blood is heated above 98.6° F. Sweating cools the body through evaporation. However, high relative humidity retards evaporation, robbing the body of its ability to cool itself.

When heat gain exceeds the level the body can remove, body temperature begins to rise, and heat related illnesses and disorders may develop.

The **Heat Index** is the temperature the body feels when heat and humidity are combined. See the below charts.

The chart below shows the **Heat Index** that corresponds to the actual air temperature and relative humidity. This chart is based upon shady, light wind conditions.

Exposure to direct sunlight can increase the **Heat Index** by up to 15° F.

Temp. (F)	Humidity Range	Heat Index Range (F)
80°	40-100%	79 - 85°
85°	40-60%	84-90°
85°	60-100%	90 - 101°
90°	40-70%	90-105°
90°	70-100%	105-121°
95°	40-50%	98-105°
95°	50 -80%	105-133°

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95°	80-100%	133°+
100°	40-70%	109- 142°
100°	70-100%	142°+
105°	40-50%	121 -133°
105°	50-100%	133 - 148°+
110°	40- 100%	135°+

Information in the above table has been paraphrased
from the National Weather Service Forecast Office Heat Index table.

Due to the nature of the heat index calculation,
the values in the tables below have an error+/- 1.3° F.

This table can be found at www.crh.noaa.gov/pub/heat.php

Heat Index Range (F)	Possible Heat Disorder
80° - 90°	Fatigue possible with prolonged exposure & physical activity.
90° - 105°	Sunstroke, heat cramps & heat exhaustion possible.
105° - 130°	Sunstroke, heat cramps, and heat exhaustion likely, and heat stroke possible.
130° or greater	Heat stroke highly likely with continued exposure.

The U.S. Environmental Protection Agency (EPA) has produced the Excessive Heat Events Guidebook (PDF, 60 pp., 1796 KB) as an additional resource for this program's information.

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5. Possible Additional Measures

The following are possible measures management may take to prevent heat illness above and beyond the foundational requirements included in this Program:

- a. Schedule maintenance and repair work in hot areas for cooler months.
- b. Schedule hot jobs for the cooler part of the day.
- c. Reduce the physical demands of workers on extreme heat days.
- d. Use relief workers or assign extra workers for physically demanding jobs.
- e. Provide cool, potable drinking water, not drinks with caffeine, alcohol or large amounts of sugar.
- f. Be aware and make employees aware that personal protective equipment (PPE) may increase the risk of heat stress. Given that PPE must be worn based upon the kind of work being performed, it is important for everyone to monitor themselves more carefully when wearing PPE and take the kind of actions provided in this Program.

Note:

This Program is kept updated based upon the most current *Heat Illness Prevention Program* implementation details on web sites such as Cal-OSHA Consultation, the Centers for Disease Control (the CDC) and the National Weather Service.

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Policy:

It is the policy of the Calimesa Fire Department to ensure the health, safety, and well-being of our employees during incidents and/or training assignments.

No employee shall be permitted to continue emergency operations or training exercises beyond safe levels of physical or mental endurance.

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Scope:

The Incident Commander / Supervisor (Company Officer) shall consider the use of REHAB on any incident or training exercise where personnel are performing arduous or extensive work or exposed to high or low ambient temperatures. Reassignment

1. Command Structure:

- A. REHAB is in the Service Branch of the Logistics Section alongside the medical Unit. On smaller incidents REHAB may work under the Operations Section or report directly to the IC. The extent of the incident dictates the location for REHAB within the command structure.
- B. There may be the need for multiple REHAB Units or Areas on an incident, The REHAB area shall be names for its geographical location.
- C. A Work/Rest crew rotation cycle will be implemented by the IC or designee based on the incident or training exercise needs.

2. Communication:

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- A. The REHAB group Supervisor will utilize the designated tactical frequency.
- B. Whenever a firefighter requires transportation for injury or illness from the REHAB Group to a hospital for further evaluation and treatment, it is the responsibility of the REHAB Group Supervisor to notify the IC or designee.
- C. REHAB may be established at any type of incident. If a firefighter requires transport for injury or illness for further evaluation and treatment from the REHAB Group during a Mass Casualty Incident (MCI) it shall be the responsibility of the REHAB Group Supervisor to ensure that MEDCOM is notified to provide the appropriate destination and transport priority.

3. Considerations:

- A. After a period of intense work, with or without wearing an SCBA, personnel will consider Informal and/or Formal-REHAB. With consideration to the following:
 - a. NWCG/OSHA Heat Index Guidelines (**Appendix A**)
 - b. IC or their designees should be aware of the cumulative effects of previous incidents and potential of future incidents on the firefighters assigned.

4. Responsibility:

- A. It shall be the responsibility of the Incident Commander / Supervisor to implement REHAB protocols.
- B. It is the responsibility of every individual to monitor his or her own exertion levels and capabilities. It is also the responsibility of every individual to comply with individual levels of adequate rest, nutrition, and hydration guidelines and recommendations.

5. Procedure:

- A. A rehabilitation Group Supervisor (Radio Designation REHAB) will be implemented to evaluate personnel for the effects of physical and mental exertion during an incident or training exercise where this Policy is implemented.
- B. Incident Commanders shall order enough resources to effectively REHAB crews assigned to incidents and training.
- C. The REHAB Group will provide a designated area(s) where personnel will assemble for:
 - a. A physical assessment
 - b. Rest, hydration, and nutrition.
 - c. Medical evaluation & continuous monitoring

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6. Forms of REHAB:

A. Informal-REHAB

- a. Informal-REHAB is the process of firefighters cooling, resting, and hydrating outside the constraints of a Formal-REHAB area. In many cases, Informal-REHAB will be the only REHAB activities performed at incidents.
- b. Informal-REHAB can have a designated location or occur at the most advantageous location to accomplish incident objectives.
- c. Informal-REHAB can include, but is not limited to, on-scene shade, in-cab AC, apparatus ice chests, and cool towels.
- d. Following Informal-REHAB, Company Officers will evaluate their crew for their ability to return to work, or the need to further REHAB.
- e. Crews and Supervisors shall adhere to the Department Policy (Section # 301.17) and N.W.C.G. Heat Illness Prevention Plan.

B. Formal-REHAB

- a. A designated REHAB Area(s), that has been established with an entry/exit point, resource tracking, medical monitoring, recovery area, and documentation.
- b. REHAB Group Location:
 - 1. Often co-located with Incident Support/Breathing Support/Medical Unit.
 - 2. Consider utilizing a bus, (School, RTA, etc.) for a large number of personnel requiring rehab in high or low ambient temperatures.
 - 3. Smoke & exhaust free environment.
 - 4. Sheltered from environment (shaded or enclosed) for the liability to cool down or warm-up
 - 5. Large enough to accommodate crews and REHAB personnel.
 - 6. Away from intrusive press and scene operations; close enough to avoid additional fatigue.

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- c. It shall be the responsibility of the REHAB Group supervisor to notify the IC when Formal-REHAB Areas have been established.
- d. It shall be the responsibility of the IC to notify the ECC and all units on-scene that Formal-REHAB has been established. Once determined Formal-REHAB is needed, an ALS ambulance will be requested to assist with REHAB and standby. A REHAB CACHE will also be requested to the incident through the City/ECC, etc.
- e. Work/Rest crew rotation will be utilized by the IC/Supervisor.
- f. Where Formal-REHAB areas have established, but geography limits their use, consider the following:
 - 1. Adjust work intensity, take frequent recovery breaks.
 - 2. Rotate crews into Formal-REHAB when possible.
- g. Adhere to NWCG/OSHA Heat Illness Prevention Plan.

7. REHAB Process:

A. When Formal-REHAB is used, the REHAB area will be staffed with sufficient personnel as dictated by the Incident Commander and must include sufficient (AMR) ALS personnel.

B. A REHAB Group Supervisor will be established to oversee the REHAB area. If incident scale dictates more than one REHAB AREA, the REHAB Group Supervisor will oversee the separate REHAB Groups. The REHAB Group Supervisor will report to the Medical Unit Leader or Logistics / Incident Commander. The REHAB Group Supervisor is responsible for:

- a. Maintaining the Resource Tracking.
- b. Ensuring that Medical Monitoring is completed.
- c. Updating Operations / Incident Commander that units have exited REHAB and are ready for re-assignment.

C. Standby Ambulance Crews along with Supervisors can be utilized to assist with staffing REHAB.

D. Medical Monitoring & Recovery

- a. All personnel entering REHAB will be assisted in removal of gear and given gross

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decontamination.

b. Personnel in REHAB will receive an initial physical assessment for:

1. Alertness and orientation.
2. Changes in gait, speech, or behavior.
3. Symptoms of heat or cold-related stress.
4. Any additional complaints.

c. personnel in REHAB will have an initial set of vital signs recorded.

1. Heat Rate
2. Blood Pressure
3. Tympanic Temperature
4. Pulse Oximetry
5. Carboxyhemoglobin (when available) 6. Glasgow Come Scale

d. Personnel will remain in the designated REHAB area until released by the REHAB personnel.

e. Hydration and passive, active, or emergency cooling or warming will be provided.

1. Passive Cooling- Occurs when natural, evaporative cooling can take place (i.e., shade, fans, removal of safety area, in-cab air conditioning).

2. Active Cooling⁰ takes place when external cooling measures are used to aid passive cooling (i.e., cold-water towels and/or submersion of forearms into ice water).

3. Emergency Cooling- rapid and aggressive cooling interventions to decrease the risk of serious complications (such as rhabdomyolysis) or death (i.e., full body cold water immersion intervention to patients as having a heat stroke).

4. Inclusion Criteria-Exertional activity of any kind with CNS compromise (A.L.O.C, unsteady gait, collapse, disoriented, headache, dizziness, etc.) and any elevated body temperature.

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5. Procedure- Remove all protective equipment and clothing from patient (underwear ok to remain in place). Place patient in ice bath while ensuring the neck and head remain above water. Continuously stir-agitate water for maximum effect. Patient should remain immersed in water for a maximum of 15 minutes and then transported. Cold water immersion interventions may be terminated early if all CNS symptoms revolve, patient begins to shiver, or if ABC's are otherwise unable to be managed.

6. Rehab personnel will monitor signs & symptoms indicating the need for further REHAB or hospital transport.

7. Prior to release from REHAB, a second set of vital signs will be taken.

f. No personnel will be allowed to resume working if while seated, their:

1. Heart Rate is greater than 120 BPM.
2. Diastolic Blood Pressure is greater than 105 mmHG.
3. Tympanic temperature is greater than 100.4F with associated CNS compromise.
4. Pulse Oximetry reading is less than 92%.
5. Carboxyhemoglobin reading is above 15ppm. . Glasgow Coma Scale of less than 15.

g. If ANY vital signs are higher than the stated thresholds after rest and hydration, it will be discretion of REHAB personnel to offer further REHAB or hospital transport. In any case, vital signs will be taken immediately prior to releasing personnel from REHAB.

h. Medical monitoring will include all ALS components when indicated.

i. Personnel who have a chief complaint onset during REHAB will be transported to the closest most appropriate hospital evaluation.

j. Any injuries or illness found during REHAB will be communicated through the chain of command.

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8. Reassignment:

- A. It is the responsibility of the REHAB Group Supervisor to track resources and make notifications when resources are available for reassignment.

9. REHAB Cache / Equipment and REHAB Locations:

The following facilities will have and maintain (2) E-Z Ups, a minimum of (15) cases of water, (15) cases of Gatorade, and (1) large ice chest.

- A. Calimesa Station # 1

City Facility (T.B.D.)

(Cross Reference Volume 2 Section 301.17 Employee Health and Safety – Heat Illness Prevention Policy and Volume 6 Section 602.11 Heat Stress Management)

Heat Index Risk and Mitigation Chart					
Reference Heat Index Chart below					
Heat Index	Risk Level	Moderate Work Mitigation		Arduous Work Mitigation	
		Rest per Hour	Water Intake per Hour	Rest per Hour	Water Intake per Hour
Very Warm 80°F - 89°F	Caution	10 min.	3/4 qt.	20 min.	3/4 qt.
Hot 90°F - 104°F	Moderate	20 min.	3/4 qt.	30 min.	1 qt.
Very Hot 105°F - 129°F	High	30 min.	3/4 qt.	40 min.	1 qt.
Extremely Hot ≥130°F	Very High to Extreme	40 min.	1 qt.	50 min.	1 qt.
				Protective Measures	
				Basic heat safety and planning	
				- Implement precautions and heighten awareness - Consider bi-modal shift - Additional precautions to protect workers - Implement bi-modal shift - Reduce production rates - Triggers additional aggressive protective measures - Implement bi-modal shift - Reduce production rates	

Moderate Work = Patrolling, hiking on level/moderate terrain, mop-up, calisthenics, etc

Hard Work = Initial Attack, fireline construction, hiking on steep terrain, endurance exercising, carrying weight, etc.

Bi-Modal Work Shift = Move essential work tasks to coolest part of operational period, and/or adjust tactics (patrol/holding, etc) during hottest part of day.

* **CAUTION:** Hourly fluid intake should not exceed 1½ qts. Daily fluid intake should not exceed 12 qts

Index Heat Chart
Temperature (°F)

	84	86	88	90	92	94	96	98	100	102	104	106	108	110
10	81	82	84	85	87	89	90	92	94	96	98	100	102	104
15	81	83	84	86	88	90	92	94	96	98	101	103	106	108
20	81	83	85	86	88	90	93	95	97	100	103	106	109	112
25	82	83	85	87	89	92	94	97	100	101	106	110	113	117
30	82	84	86	88	90	93	96	99	102	106	110	114	118	122
35	83	85	87	89	92	95	99	102	106	110	114	119	124	129
40	83	85	88	91	94	97	101	105	109	114	119	124	130	136
45	84	87	90	93	96	100	104	109	114	119	125	131	136	144
50	85	88	91	95	99	103	108	113	118	124	131	137	144	152
55	86	89	93	97	102	107	112	117	124	131	138	145	154	162
60	88	91	95	100	105	110	116	123	129	137	145	153	162	171
65	89	93	98	103	107	115	121	128	136	145	153	162	172	
70	90	95	100	106	112	119	126	134	143	152	161	172	182	
75	93	98	104	110	117	124	132	141	151	160	172	180		
80	94	100	106	113	121	129	138	148	158	169	181			

% Relative Humidity

- Exertion level of training or work output, especially on several sequential days
- Acclimatization (and other individual risk factors – see reverse side)
- Time of exposure, duration of exposure will determine level of mitigation

Remove contaminated clothing and remove gloves during rest periods when from hazards on the away

Ensure clothing is clean to allow dissipation of heat and sweat from Consider removing line pack when performing non-critical fireline tasks following a assessment to mitigate risk exposure

Monitor personnel for hydration and food

Encourage frequent drinking of water and sports drinks) Do not exceed 1½ quarts per hour or 12 quarts per day

Spot checks by supervisor, and assigned Cache water or buddy

Ensure firefighters hydrate before, during, and after work

Encourage stacking throughout the work

Table salt may be added to food or eat more salty foods when the heat high (very hot/extremely hot) category is

Salt tablets are not

Mitigation(s)

Determine heat index hourly through weather

Adjust work/rest cycles according to Heat Index Risk and Mitigation

Adjust operational tactics to avoid hottest part of day (e.g., bi-modal or production). Set up buddy system and have firefighters to watch for signs of Ensure emergency medical response plan is in place, and known to Reschedule non-mission critical tasks when HI is



Individual Risk Factors:

The more factors, the higher the risk

- Sources of Heat**
- Internal heat generated by physical labor
- Environmental conditions**
- High heat environments can restrict the body's ability to evaporate sweating and to cool itself
 - Direct sunshine can increase heat index values up to 15°F
- Not acclimatized**
- Initial exposure (<4-5 days) of any of the following:
 - Increased heat exposure
 - Increased exertion levels of work/training
 - Lack of quality sleep

- Lower fitness level** requires more acclimatization time.
- Overweight**
- Minor illness**
- Taking medications**, either prescribed or over-the-counter and/or dietary aids. Example: Allergy or cold remedies, certain antibiotics, statins (cholesterol lowering drugs), ephedra and creatine
- Dehydration** occurs during prolonged exertion when fluid loss is greater than fluid intake.

- Excessive caffeine intake**
- Supplements** are not regulated and have potential to impact firefighters.
- Prior history of heat illness** – a heat stroke event at any point, or an episode of lesser HRI in previous 24-48 hours
- Skin disorders** such as heat rash and sunburn

The intent of this guide is to provide a concise collection of Heat Related Illness (HRI) prevention information for operational field use. Its contents are not absolute rules and HRI potential and prevention mitigations should be evaluated and balanced with all the risks associated with wildland fire operations.

This Guide includes:

- Hazards
 - Individual Risk Factors
 - Fluid Replacement and Work/Rest Guide
 - Heat Illness Mitigation
 - Warning Signs and Symptoms of HRI
- Hazards:**
- Heat Index, high ambient temperature and relative humidity

that prevent effective sweating

Download the OSHA Heat Safety Tool App. free
online through your application store.

Radiant Heat = sun/flame can increase HI
values by up to 15°



Warning Signs and Symptoms of Heat Stress and Illness

Indications of Possible Heat Related Illness

COMMON SIGNS/SYMTOMS	IMMEDIATE ACTIONS
• Dizziness • Headache • Nausea • Unsteady walk • Weakness or fatigue • Muscle Cramps	• Stop current activity • Allow person to rest in shade • Take sips of water • Call for medical assistance to evaluate individual
SERIOUS SIGNS/SYMTOMS	If no emergency medical personnel are available, call for ambulance or medical transport. Immediately implement medical emergency plan and call for an ambulance and/or medivac for emergency transport while doing the following: • Lay individual down in shade until MEDEVAC or ambulance arrives • Remove clothing from individual • Apply ice packs, or move individual to vehicle with air conditioning • Pour or immerse body in cold water and fan • Give sips of water while awaiting ambulance (if conscious) • Monitor airway and breathing until ambulance or MEDEVAC arrive • Continue cooling during transport or until body reaches 100° F

Warning Signs and Symptoms of Heat Stress and Illness

An important sign that the firefighter is in a **serious life-threatening** condition is an altered mental status (with or without increased temperature).

Ask the following questions:

- **What is your name?** (Does not know their name.)
- **What month is it? What year is it?** (Does not know the month or year.)
- **Where are we/you?** (Is not aware of location or surroundings.)
- **What were you doing before you became ill?** (Does not know the events that led to the present situation.)

Call for emergency MEDEVAC or ambulance

Indications of Possible Water Intoxication (Over-Hydration)

Water intoxication (hyponatremia) occurs when an individual drinks excessive amounts of water and decreases the sodium to water concentration in the body.

Signs and Symptoms: Confusion, Weakness, and Vomiting

For a full reference list for the HIP Pocket Guide, go to: <http://www.nwccg.gov/branches/pre/rmc/index.htm>



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PURPOSE

The purpose of the Exposure Control Plan for Bloodborne Pathogens policy is to provide a comprehensive infection control system of protection against infectious diseases for the Fire Department members. The Exposure Control Plan for Bloodborne Pathogens Policy entails the following:

- i. Authority
- ii. Objectives
- iii. Exposure Determination
- iv. Responsibilities/Duties
- v. Plan Definitions
- vi. Availability of the Exposure Control Plan to Personnel
- vii. Review and Update of the Plan
- viii. Health Maintenance
- ix. Infection Control Training
- x. Decontamination
- xi. Station Environment
- xii. Personal Protective Equipment
- xiii. Scene Operations
- xiv. Post -Response
- xv. Post Exposure Protocol
- xvi. Area Hospitals Point Of Contact For Requests A Communicable Disease Determination

I. AUTHORITY

- A. The Exposure Control Plan for Bloodborne Pathogens policy meets or exceeds the following:

- i. FEDERAL OCCUPATIONAL SAFETY AND HEALTH ADMINISTRATION (FED –OSHA 29 CFR 1910.1030 & 29 CFR 1910.1920)
 - ii. CALIFORNIA OCCUPATIONAL SAFETY AND HEALTH ADMINISTRATION (C.A.L.- OSHA TITLE 8, Chapter 4, SECTION 5193)

II. OBJECTIVES

- A. Identify tasks and corresponding job classifications that are high risk for exposure to blood, other body fluids, or potentially infectious materials may occur.

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- B. Establish a schedule for implementation of the Fire Department's Exposure Control Plan for Bloodborne Pathogens Policy.
- C. Identify the procedure for the evaluation of circumstances surrounding exposure incidents.
- D. Require all personnel to adhere to the established policy when interacting with a communicable disease or a suspected infectious disease.

III. EXPOSURE DETERMINATION- TASKS and JOB CLASSIFICATION

- A. The following tasks are high risks to exposure to blood, body fluids, or other potentially infectious materials:
 - i. Conducting patient assessment
 - ii. Providing supplemental oxygen therapy
 - iii. Maintaining or providing airway control
 - iv. Suctioning the airway
 - v. Obtaining intravenous access
 - vi. Orotracheal nasotracheal intubation
 - vii. Orotracheal nasotracheal suctioning
 - viii. Orogastic suctioning
 - ix. Obtaining blood samples
 - x. Obtaining intraosseous access
 - xi. Placing oropharyngeal devices
 - xii. Placing nasopharyngeal devices
 - xiii. Inserting orogastric or nasogastric tubes
 - xiv. Pleural decompression